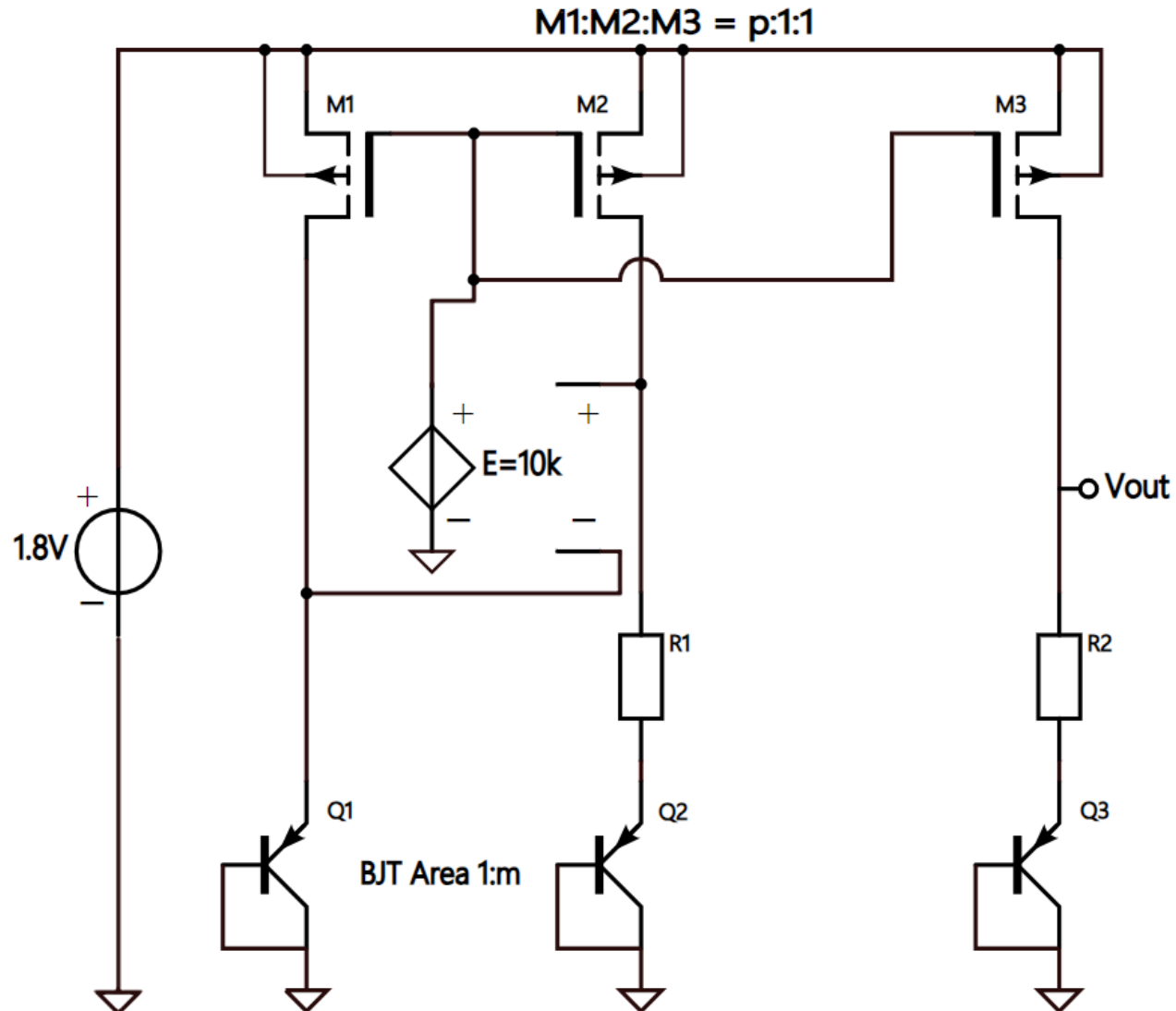


HW



- Open file HW1_bandgap
- For the bandgap circuit shown, calculate the value of R1 such that the emitter current of Q1 is 1uA at room temperature. Assume that M1 and M2 form an **ideal** current mirror ($p=1$), and set the emitter-area ratio $m = 5 +$ the first digit of **your** student number, **(5%)**.
 - Then calculate the value of R2 such that Vout is a band-gap voltage that is maximally flat at room temperature **(10%)**.
 - Now using real PMOS devices, find resistor values (R1, R2) and transistor sizes (M1-M3) such that the spread in Vout is $< 1\%$ over process (TT, SS, FF), supply voltage V_{DD} (1.6V to 1.8V) and temperature (-40 to 85°C). Submit the resulting parameters and plots of Vout vs temp over process and VDD **(15%)**.

HW

- d) Simulate the noise (include the plots) of the bandgap circuit you designed in (c). Which device is the dominant contributor to the noise in V_{out} ? Optimize the design of the PMOS current sources to minimize their noise contribution, while still meeting the PVT specifications given in c) **(10%)**.
- e) Design an amplifier (include the schematic) to replace the ideal VCVS shown in the figure **(20%)**.
 - a. Its contribution to the thermal noise density of V_{out} should be less than the combined noise contribution of Q1, Q2 and R1.
 - b. Its DC gain should be as high as possible, but at least $> 40\text{dB}$ (SS corner, -40°C and 1.6V)
 - c. It may be **biased** by ideal current and voltage sources.
 - d. Its power dissipation (excluding the bias circuits) should be $< 16\mu\text{W}$
- f) In the bandgap circuit you designed in (c), replace the ideal VCVS with your own amplifier. Fine-tune the amplifier (include the schematics) to ensure that the resulting V_{out} still meets the PVT specifications given in (c) **(20%)**.
- g) Add 1mV of offset to the input of the amplifier. What is the change in V_{out} ? The impact of the offset can be dramatically improved by chopping. Demonstrate this by adding ideal choppers to your amplifier and computing the average value of V_{out} over two chopper phases (specify the residual error). For your design, what would be a good choice of the chopping frequency? Motivate your answer **(20%)**.

Note: The minimum dimensions of all MOSFETs used should be $> 1\mu\text{m}$.

LT Spice library (works in LTSpice V24)

- Use **log018_mod_V24.l** from Brightspace, log018_mod.l does not work in version 24 of Ltspice, only in V17!
- For the bipolar simulations over corners, you have to use the 'log018_mod.l'
- For simulating the BJTs over corners, use SS_BIP, FF_BIP etc.
- For simulating mosfets over corners, use SS, FF etc.

```
.lib '/Users/florisvanmouri/Library/CloudStorage/OneDrive-DelftUniversityofTechnology/PhD/Education/Analog_Fund/log018_mod.l' SS_BIP
.lib '/Users/florisvanmouri/Library/CloudStorage/OneDrive-DelftUniversityofTechnology/PhD/Education/Analog_Fund/log018_mod.l' SS
```

- Before using it, make sure that you changed all **26** path directories in the 'log018_mod.l' file to the directory of the file. For instance:

```
*
*****
*****
*
*          1.8V NORMAL DEVICES LIB
*
*****
***** CORNER_LIB OF TYPICAL MODEL *****
.LIB TT
.param
+toxnp = 4.08E-09      toxpp = 4.08E-09
+dvthnp = 0           dvthpp = 0
+dxl = 0              dxw = 0
+cjnp = 0.001000266   cjp = 0.001121
+cjswnp = 2.040547E-10  cjswpp = 2.481E-10
+cjswgn = 3.340547E-10  cjswgp = 4.221E-10
+cgon = 3.665E-10      cgop = 3.28E-10
+ndfnp = 2E-07         ndfpp = 2E-07
.lib '/Users/florisvanmouri/Library/CloudStorage/OneDrive-DelftUniversityofTechnology/PhD/
Education/Analog_Fund/log018_mod.l' MOS
.ENDL TT
***** CORNER_LIB OF SNSP MODEL *****
.LIB SS
.param
+toxnp = 4.213E-09      toxpp = 4.213E-09
+dvthnp = 0.1          dvthpp = -0.067
+dxl = 1.33E-08        dxw = -2.2E-08
+cjnp = 0.001050279    cjp = 0.00117705
+cjswnp = 2.142574E-10  cjswpp = 2.60505E-10
```